Worksheet Title

Subtitle or topic area

Opening description goes here.

1. **First kernel fact.** A short, memorable statement.
2. **Second kernel fact.** Another one.

Kernel — The minimal set of facts you commit to memory; everything else is derived from them rather than memorised separately.

Divide both sides by $\cos^2(\theta)$

Starting from $\sin^2(\theta) + \cos^2(\theta) = 1$, divide every term by $\cos^2(\theta)$:

$$\frac{\sin^2(\theta)}{\cos^2(\theta)} + \frac{\cos^2(\theta)}{\cos^2(\theta)} = \frac{1}{\cos^2(\theta)}$$
$$\tan^2(\theta) + 1 = \sec^2(\theta)$$

In practice

This identity lets you integrate $\tan^2 x$ by rewriting it as $\sec^2 x - 1$, which is on the standard antiderivative list.

In signal processing

A specific title can be passed when the application is concrete enough to name.

Now derive $1 + \cot^2(\theta) = \csc^2(\theta)$ the same way.

1. Practice Problems

Problem 1 A straightforward problem to build confidence.

Hint: This is a hint, it's meant to be difficult to read.

Problem 2 A problem requiring two or three steps.

Problem 3 A harder problem that requires connecting two ideas from this worksheet.

(a) Try this (b) Then try this (c) Then try this

The derivative of $\sin(x)$ is ____________, and the derivative of $\cos(x)$ is ____________.

Answers

Answer 1
Answer to the problem would go here.

Answer 3
(a) 100

(b) 200

(c) 300